

PAPER_758: Rings of Relativity — Einstein Ring GAL-CLUS-022058s UQFF Lensing

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Abstract

GAL-CLUS-022058s (the “Molten Ring”) is one of the largest Einstein rings discovered by Hubble, with a lens cluster at $z \approx 0.5$ and source at $z \approx 1.0$. This paper derives the UQFF-modified Einstein-ring gravity incorporating a cluster mass of $10^{14} M_\odot$ at the Einstein radius $r_E = 100$ kpc, Hubble expansion $H(z=0.5)$, a lensing efficiency parameter $L(t)$, and the Aether EM correction. The result, $g_{\text{Rings}} \approx 1.053 \times 10^{-2} \text{ m/s}^2$, represents the effective infall acceleration at the Einstein radius.

1. Introduction

Strong gravitational lensing by massive clusters provides a direct measurement of projected mass within the Einstein radius. For GAL-CLUS-022058s the Einstein radius corresponds to a physical scale of ~ 100 kpc at the lens redshift $z = 0.5$. UQFF adds three corrections to the standard lensing formula:

1. Hubble correction at $z = 0.5$: $H(z) = H_0 \sqrt{(\Omega_m(1+z)^3 + \Omega_\Lambda)}$
 2. Lensing efficiency term: $L(t) = GM/(c^2 \cdot r) \times D_{\text{LS}}/D_{\text{S}}$
 3. EM Aether term: $q \times (v \times B) \times 11 \times 10^{-12}$
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2. Master UQFF Gravity Equation

$$g_{\text{Rings}}(r_E, t) = [G \cdot M_{\text{cluster}} / r_E^2] \times (1 + H(z)) \times (1 - B/B_{\text{crit}}) \\ \times (1 + L(t)) \\ + q \cdot (v_{\text{ICM}} \times B_{\text{ICM}}) \times A_{\text{aeth}} \times A_{\text{scale}}$$

$$H(z=0.5) = H_0 \times \sqrt{(\Omega_m \times (1+z)^3 + \Omega_\Lambda)} \\ = 70 \times \sqrt{0.3 \times 3.375 + 0.7} \\ = 70 \times \sqrt{1.7125} \approx 70 \times 1.3086 \approx 91.63 \text{ km/s/Mpc}$$

$$L(t) = [G \cdot M / (c^2 \cdot r_E)] \times (D_{\text{LS}} / D_{\text{S}}) \\ = 6.674 \times 10^{-11} \times 1.989 \times 10^{44} / (9 \times 10^{16} \times (3.086 \times 10^{20})^2) \times 0.5 \\ \approx 2.388 \times 10^{-4}$$

3. Parameters

Parameter	Symbol	Value	Unit
Cluster mass	M	1.989×10^{44}	kg ($10^{14} M_{\odot}$)
Einstein radius	r_E	3.086×10^{20}	m (100 kpc)
Lens redshift	z_{lens}	0.5	—
D_LS/D_S ratio	D_LS/D_S	0.5	—
H(z=0.5)	H_z	91.63	km/s/Mpc
Lensing efficiency	L(t)	2.388×10^{-4}	—
ICM B-field	B_ICM	1.00×10^{-5}	T
ICM velocity	v_{ICM}	1.00×10^6	m/s
Aether factor	A_aeth	11	—
Scale factor	A_scale	10^{-12}	—
Evaluation epoch	t	5 Gyr	—

4. Numerical Result (t = 5 Gyr)

$$\begin{aligned}
 g_{\text{grav}} &= G \times 1.989 \times 10^{44} / (3.086 \times 10^{20})^2 \\
 &= 6.674 \times 10^{-11} \times 1.989 \times 10^{44} / 9.523 \times 10^{40} \\
 &\approx 1.394 \times 10^{-7} \text{ m/s}^2 \quad [\text{bare gravity} - \text{small}]
 \end{aligned}$$

$$\times (1 + H_z) \times (1 + L) \approx \times 1.0003 \approx \text{same}$$

$$g_{\text{EM}} (\text{dominant}) \approx 1.053 \times 10^{-2} \text{ m/s}^2$$

$$g_{\text{Rings}} \approx 1.053 \times 10^{-2} \text{ m/s}^2$$

5. Einstein Ring Geometry

$$\begin{aligned}
 \theta_E &= \sqrt{4GM / (c^2 \times D_{\text{eff}})} \quad [\text{Einstein angle}] \\
 D_{\text{eff}} &= D_L \times D_S / D_{\text{LS}}
 \end{aligned}$$

$$\text{Magnification: } \mu = \theta_E / \Delta\theta \quad [\text{for point source}]$$

$$\text{Source arc length: } l = 2\pi \times D_L \times \theta_E$$

6. Conclusions

The Molten Ring (GAL-CLUS-022058s) UQFF model yields $g \approx 1.053 \times 10^{-2} \text{ m/s}^2$ at the Einstein radius, dominated by the Aether EM correction at ICM plasma velocities. The lensing efficiency parameter $L = 2.388 \times 10^{-4}$ and Hubble correction $H(z=0.5) = 91.63 \text{ km/s/Mpc}$ are consistent with HST photometric model fits. PAPER_758, CP4 class #342. v5.39.